INTELLITWIN: AI-DRIVEN DIGITAL TWIN FRAMEWORK FOR PREDICTIVE MAINTENANCE

B.Tech Project Synopsis — Phase 3: Problem Definition, Research Gap & Research Questions

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3. Problem Definition, Research Gap and Research Questions

3.1 Problem Statement

In modern industrial systems and mission-critical cyber-physical production environments, rotary engineering assets—most notably commercial aviation turbofan engines, industrial gas turbines, and high-load transmission assemblies—operate under relentless thermal gradients, mechanical fatigue, and variable aerodynamic stresses. Sustaining high operational availability while precluding in-service catastrophic failure constitutes the primary objective of Prognostics and Health Management (PHM) and Predictive Maintenance (PdM) [1], [2].

Formally, consider a fleet of industrial assets indexed by $i \in \{1, 2, ..., N\}$. Each asset undergoes continuous physical degradation across discrete operational time cycles $t \in \{1, 2, ..., T_i\}$, where $T_i \in \mathbb{N}$ denotes the true, unobserved end-of-life cycle at which the machine suffers functional failure. At any operational cycle t, the true Remaining Useful Life (RUL) of asset i is defined as the remaining operational duration prior to terminal failure:

$$y_i(t) = T_i - t \qquad (1)$$

At each cycle t, onboard condition-monitoring sensors record a multivariate telemetry observation vector $\mathbf{x}_i(t) \in \mathbb{R}^{d_x}$ alongside external operating regime descriptors $\mathbf{s}_i(t) \in \mathbb{R}^{d_s}$ (such as altitude, Mach number, and throttle settings) [1]. To ensure physical causality and prevent future-state data leakage, prognostic predictions at cycle t must depend strictly upon the causal historical trajectory up to time t, encapsulated by a fixed sliding window of length W:

$$\boldsymbol{X}_i(t) = [\boldsymbol{z}_i(t - W + 1), \boldsymbol{z}_i(t - W + 2), ..., \boldsymbol{z}_i(t)] \in \mathbb{R}^{W \times (d_x + d_s)} \qquad (2)$$

where $\mathbf{z}_i(t) = [\mathbf{x}_i(t), \mathbf{s}_i(t)]$ denotes the combined condition-setting feature vector at cycle t.

While standard machine learning approaches train a deterministic regression function f: $\mathbb{R}^{W \times (d_x + d_s)} \rightarrow \mathbb{R}_+$ to generate scalar point estimates $\hat{y}_i(t) = f(\mathbf{X}_i(t))$, point predictions fail to capture aleatoric sensor noise and epistemic model uncertainty [3], [4]. In safety-critical industrial operations, predictive maintenance requires calibrated prediction intervals:

$$C_{1-\alpha}(\boldsymbol{X}_i(t)) = [L_i(t), U_i(t)] \qquad (3)$$

targeting a pre-specified nominal marginal coverage probability $1 - \alpha \in (0, 1)$, such that:

$$\mathbb{P}(y_i(t) \in [L_i(t), U_i(t)]) \geq 1 - \alpha \qquad (4)$$

Crucially, predictive maintenance is not an isolated statistical estimation task; rather, prognostic intervals directly drive a sequential operational stopping time $\tau_i \in \{W, W + 1, ...\}$ at which the asset is retired for overhaul:

$$\tau_i = inf\{ t \geq W : \pi(C_{1-\alpha}(\boldsymbol{X}_i(t))) = MAINTAIN \} \qquad (5)$$

where π represents the operational maintenance policy. The resulting lifecycle economic cost incurred by asset i is governed by severe asymmetric penalties:

$$C_i(\tau_i) = C_{PM} \cdot \&DoubleStruckOne;(\tau_i < T_i) + C_{FAIL} \cdot \&DoubleStruckOne;(\tau_i \geq T_i) + C_{WASTE} \cdot (T_i - \tau_i) \cdot \&DoubleStruckOne;(\tau_i < T_i) \qquad (6)$$

where C_{PM} represents the baseline cost of a scheduled preventive replacement, C_{FAIL} is the severe penalty associated with an unscheduled, catastrophic in-service breakdown ($C_{\mathrm{FAIL}} \gg C_{\mathrm{PM}}$), and C_{WASTE} denotes the economic penalty per cycle of unutilized, discarded useful life [5], [6]. The central problem investigated by IntelliTwin is formulating an uncertainty-aware framework that connects condition-adaptive conformal RUL prediction intervals directly with this sequential stopping decision, systematically evaluating whether decision-aware calibration can enhance maintenance utility while preserving rigorous coverage guarantees [3], [7].

3.2 Limitations of Existing Approaches

A systematic audit of the contemporary prognostic and predictive maintenance literature reveals four major structural limitations in existing approaches:

1. **Pervasive Reliance on Uncalibrated Point Predictions:** The overwhelming majority of deep learning and ensemble RUL models benchmark performance exclusively on symmetric loss metrics such as Mean Squared Error (MSE) or

Root Mean Squared Error (RMSE) [1], [8]. Point estimators implicitly treat every prediction with uniform certainty, providing no probabilistic margin or tail risk quantification [3], [4]. In real-world asset management, overestimating RUL by 10 cycles produces catastrophic in-flight failure, whereas underestimating RUL by 10 cycles merely causes premature overhaul [1]. Point predictors are fundamentally incapable of communicating this operational asymmetry to plant decision-makers.

2. **Conditional Miscalibration and Subgroup Vulnerability:** Conventional uncertainty quantification methods—including Gaussian Process regression, Monte Carlo dropout, and standard Inductive Conformal Prediction (ICP)—predominantly assume exchangeable or independent and identically distributed (i.i.d.) observations [4], [9]. However, machine degradation telemetry violates exchangeability due to profound temporal autocorrelation and operating regime transitions (e.g., multi-altitude flight envelopes in C-MAPSS FD002) [1], [3]. While global conformal calibration guarantees marginal coverage over the entire population, recent studies indicate that it frequently conceals severe conditional miscalibration, suffering systematic undercoverage during high-stress operating regimes or critical late-life degradation stages ($t \to T_i$) [3], [10]. An engine approaching its end-of-life may experience true coverage substantially below the nominal $1 - \alpha$ target, creating hidden catastrophic operational hazards.
3. **Decoupling of Statistical Uncertainty from Downstream Decision Economics:** In published prognostic research, uncertainty quantification and maintenance decision optimization are almost universally formulated as disjoint, uncoupled problems [6], [7]. Statistical calibrators are typically optimized to minimize symmetric interval width (MPIW) or Winkler interval loss [11], without accounting for the fact that downstream maintenance policies trigger replacement based predominantly on the conservative lower prediction bound $L_i(t)$ [6], [7]. Consequently, an interval calibration that appears statistically optimal under symmetric loss may trigger excessive false alarms in low-risk operational states or fail to provide adequate conservatism during critical wear phases.
4. **Conflation of Genuine Decision Value with Trivial Over-Conservatism:** When maintenance policies employ lower confidence bounds to schedule overhauls, naive baselines—such as uniformly widening prediction intervals, adding ad-hoc point-prediction safety margins, or systematically shifting maintenance triggers to earlier flight cycles—can artificially eliminate in-service failures [6], [12]. However, this failure avoidance is achieved through extreme conservatism that squanders immense usable component life (C_{WASTE}) and inflates overall fleet

lifecycle expenditure [5], [6]. Current literature lacks rigorous experimental controls to disambiguate whether observed maintenance cost reductions stem from informative, decision-aware uncertainty calibration or merely reflect trivial, over-conservative early replacement policies.

Crucially, these limitations do not imply that all existing techniques fail simultaneously; rather, prior literature addresses these challenges in fragmented silos without a cohesive framework that unifies conditional calibration with downstream sequential maintenance utility [3], [6], [7].

3.3 Research Gap

Synthesizing the state-of-the-art across prognostic health management, conformal prediction, and maintenance optimization reveals a distinct research gap:

> *"Existing RUL research has separately studied conformal uncertainty calibration, condition/regime-aware calibration, and uncertainty-aware maintenance decision-making. What remains insufficiently established is a RUL-specific method that jointly adapts calibration to operating conditions and downstream sequential maintenance consequences while preserving reliable uncertainty coverage. Our study will therefore investigate whether decision-aware, condition-adaptive conformal calibration can improve maintenance utility without sacrificing subgroup reliability or merely becoming more conservative."*

The boundaries of this research gap are delineated relative to verified prior art:

- Javanmardi and Hüllermeier (2023) [4] evaluated split conformal prediction on C-MAPSS turbofan data, but focused strictly on marginal coverage without condition-adaptive scaling or downstream maintenance decision evaluation.
- Diao et al. (2026) [3] integrated conformalized quantile regression (CQR) with LSTM networks on C-MAPSS FD001 and FD002, conducting life-stage analysis and target adaptation, but explicitly positioned their methodology upstream of prescriptive maintenance optimization.
- Zhu et al. (2025) [5] and Walia and Kumar (2026) [8] modeled maintenance-cost optimization and reinforcement learning scheduling on aero-engine telemetry, but utilized parametric distributions or quantile heuristics that lack finite-sample, distribution-free coverage guarantees.
- Robinson (2026) [10] developed asymmetric CQR prediction intervals across C-MAPSS subsets, but did not incorporate hierarchical engine-level weighting or

benchmark against prespecified matched-conservatism baselines.

- Cortes-Gomez et al. (2025) [7] introduced utility-directed conformal prediction in generic machine learning benchmarks, yet its formulation has not been investigated or established for the sequential, non-exchangeable degradation dynamics inherent to predictive maintenance digital twins.

IntelliTwin does not assert absolute novelty for conformal prediction, deep prognostic modeling, or maintenance cost simulation in isolation; rather, the project investigates their method-level integration within a rigorous, pre-registered empirical framework.

3.4 Proposed Research Questions

To investigate the identified research gap, the IntelliTwin research design establishes four formal, pre-registered research questions (RQ1–RQ4). In strict accordance with the project's frozen research protocol (ADR-018), the exact approved wording is preserved:

Research Question 1 (RQ1)

Can a decision-aware, condition-adaptive conformal RUL calibration method improve sequential maintenance utility while preserving reliable prediction-interval coverage?

- **Investigation & Experiment Mapping:** Evaluated via **Experiment A**, benchmarking the proposed decision-aware, condition-adaptive conformal calibration method against global conformal calibration and standard condition-aware calibration on C-MAPSS FD002 under the primary reference scenario ($C_{\text{PM}} = 1.0$, $C_{\text{FAIL}} = 5.0$, $C_{\text{WASTE}} = 0.20$) at 90% nominal coverage [1], [6].
- **Evaluation Criteria:** Primary outcome is total simulated maintenance cost contrast $\Delta_{\text{cost}} = \text{Cost}_B - \text{Cost}_P > 0.20$ Standardized Cost Units (SCU) per engine over both baselines, subject to marginal and subgroup absolute coverage error ceilings $E < 0.05$ and non-increasing fleet failure rates.

Research Question 2 (RQ2)

Can the proposed method improve reliability across operating regimes and degradation conditions without excessively widening prediction intervals?

- **Investigation & Experiment Mapping:** Evaluated via **Experiment B**, assessing conditional coverage and worst-group reliability across six discrete operating regimes and four degradation lifecycle quartiles on FD002 [1], [3].
- **Evaluation Criteria:** Primary outcome is worst-group conditional coverage error contrast $\Delta_{\text{error}} = \text{WorstErr}_B - \text{WorstErr}_P > 0.01$ (1 percentage point), while

restricting Mean Prediction Interval Width (MPIW) and Winkler interval score deterioration to under 1.0 cycle ($D_{\text{width}} < 1.0$, $D_{\text{score}} < 1.0$) [11].

Research Question 3 (RQ3)

Are the maintenance benefits of the proposed method attributable to decision-aware uncertainty calibration rather than simply more conservative prediction intervals or earlier maintenance?

- **Investigation & Experiment Mapping:** Evaluated via **Experiment C**, disambiguating true calibration decision value from naive conservatism by benchmarking against three prespecified matched-conservatism control families: widened prediction intervals, tuned point-prediction safety margins, and earlier-intervention heuristic policies [6].
- **Evaluation Criteria:** Primary outcome is achieving maintenance cost contrast $\Delta_{\text{cost}} > 0.20$ SCU per engine against *every* prespecified matched-conservatism control family while preserving coverage constraints ($E < 0.05$).

Research Question 4 (RQ4)

Does the proposed method retain its calibration and maintenance-decision benefits across operating conditions, base RUL predictors, datasets, and repeated experimental runs?

- **Investigation & Experiment Mapping:** Evaluated via **Experiment D**, testing generalizability across a mandatory 6-cell configuration matrix: FD001 (single regime), FD002 (multi-regime), and FD004 (multi-regime, multi-fault) across both Gradient Boosted Regressors (GBR) and Long Short-Term Memory (LSTM) neural networks across five random seeds (2026–2030), weighted equally (1/6 each) [1].
- **Evaluation Criteria:** Primary outcome is achieving weighted aggregate improvements ($\Delta_{\text{cost}} > 0.20$ SCU, $\Delta_{\text{error}} > 0.01$) without exhibiting material harm ($\Delta_{\text{cost}} < -0.20$ SCU, $\Delta_{\text{error}} < -0.01$) in any individual configuration cell.

3.5 Research Significance

The research significance of the IntelliTwin initiative resides in establishing a mathematically principled, economically grounded methodology that bridges statistical uncertainty quantification with physical asset management:

1. **Methodological Grounding:** Conformal prediction provides finite-sample, distribution-free marginal coverage guarantees without relying on restrictive Gaussian or asymptotic assumptions [4], [9]. IntelliTwin extends this framework

to address the temporal autocorrelation and operating regime shifts characteristic of turbofan engines, formulating engine-level cluster bootstrapping and family-wise error rate control (Bonferroni $\alpha = 0.05 / 4 = 0.0125$) to preclude false discoveries [13].

2. **Economic Alignment via Standardized Cost Units:** Rather than evaluating prognostic algorithms solely on abstract statistical loss functions, IntelliTwin evaluates decision policies within a discrete-event sequential simulator governed by Standardized Cost Units (SCU) [5], [6]. This ensures that prediction intervals are judged directly by their operational efficacy in balancing planned maintenance against catastrophic breakdown.
3. **Open-Science Governance and Demarcation:** In strict compliance with academic research integrity, all research questions, hypotheses, and decision criteria are pre-registered prior to confirmatory evaluation. The project explicitly distinguishes implemented engineering pipelines from experimental questions under empirical investigation. By establishing formal criteria for both supportive and non-supportive outcomes, IntelliTwin ensures that its empirical findings deliver unbiased, reproducible scientific value to the engineering community.

3.6 Problem Scope and Boundaries

To maintain academic rigor and engineering precision, the experimental boundaries of the IntelliTwin framework are explicitly delineated based on repository evidence:

- **Target Benchmark Assets:** The study evaluates turbofan engine degradation utilizing the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) benchmark suite [1]:
  - *FD001:* Single operating condition (Sea Level), single failure mode (High-Pressure Compressor degradation); 100 training engines, 100 test engines. Used as baseline control.
  - *FD002:* Six operating conditions, single failure mode (HPC degradation); 260 training engines, 259 test engines. Primary benchmark for multi-regime calibration.
  - *FD004:* Six operating conditions, two failure modes (HPC and Fan degradation); 249 training engines, 248 test engines. Primary benchmark for complex multi-fault robustness.
- **Feature Representation:** In accordance with the project's verified schema, 6 invariant/non-informative sensor channels (indices 1, 5, 10, 16, 18, 19) are removed, leaving 14 active degradation sensor measurements alongside 3 operational setting descriptors (17 features total). Sensor values are standardized

via training-only Z-score normalization (μ, σ fitted solely on training splits with ddof = 0).

- **Causal Windowing Protocol:** Telemetry is processed into strictly causal backward sliding windows of length $W = 30$ cycles, completely barring future-cycle information.
- **Base Prognostic Models:** Evaluates two distinct supervised learning paradigms: Histogram Gradient Boosting Regressors (`HistGradientBoostingRegressor`) over flattened $30 \times 17 = 510$-dimensional vectors, and unidirectional PyTorch Long Short-Term Memory (`LSTMRegressor`) networks over 3D sequence tensors (B, 30, 17) [1], [8].
- **Decision Simulator Parameters:** Evaluated within a discrete-event sequential maintenance simulator under Standardized Cost Units ($C_{PM} = 1.0$, $C_{FAIL} = 5.0$, $C_{WASTE} = 0.20$ per cycle) [6].
- **Explicit Scope Exclusions:**
  - *No Live Physical Sensor Ingestion:* The framework does not interface with live edge IoT hardware, physical sensor transceivers, or industrial fieldbus protocols (e.g., OPC UA, MQTT, Modbus).
  - *No Physical Closed-Loop Actuation:* The framework does not transmit physical actuation signals or feedback control back into operational machinery.
  - *Digital Twin Scope:* The Digital Twin is implemented as an offline, trace-driven simulation replay engine (`TwinReplayEngine` and FastAPI web service at port 8000) that streams historical C-MAPSS telemetry cycle-by-cycle at verified numerical parity (atol $\leq 10^{-4}$), providing real-time prognostic dashboard visualization without physical back-coupling [2].

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